

VF10

Consider the unidirectional periodic motion induced in an infinitely long circular pipe of radius a by an oscillatory pressure gradient $P_l = A \cos(\omega t)$. Write the conservation equation with boundary conditions that determine the axial velocity $v_x(r, t)$ and show how the problem can be solved exactly by separation of variables. Study separately the limits $a^2 \omega / \nu \gg 1$ and $a^2 \omega / \nu \ll 1$ and obtain the corresponding limiting solutions. In biofluid mechanics the square root $a(\omega/\nu)^{1/2}$ is called the Womersley parameter, which takes fairly large values for blood flow in large arteries, as you can see by using the values corresponding to the human aorta ($\mu/\rho \simeq 4 \times 10^{-2} \text{ cm}^2/\text{s}$, $a \simeq 1.2 \text{ cm}$, and $\omega = 2\pi \text{ s}^{-1}$), but that decreases for flow in smaller arteries. Find how small the artery radius needs to be for Poiseuille flow to be approximately applicable.

$$\rho \frac{\partial v_x}{\partial t} = A \cos(\omega t) + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) \quad \begin{cases} r=0: \frac{\partial v_x}{\partial r} = 0 \\ r=a: v_x = 0 \end{cases}$$

$\rho \nu \omega / a^2 \quad A \quad \mu \nu / a^2$

EXACT SOLUTION:

$$\rho \frac{\partial v^*}{\partial t} = A e^{i\omega t} + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v^*}{\partial r} \right) \quad \begin{cases} r=0: \frac{\partial v^*}{\partial r} = 0 \\ r=a: v^* = 0 \end{cases}$$

$$v^* = A e^{i\omega t} f(r) \rightarrow \rho i\omega f = 1 + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) \quad \begin{cases} \frac{\partial f}{\partial r} = 0 \text{ at } r=0 \\ f = 0 \text{ at } r=a \end{cases}$$

$$\frac{O(\rho \frac{\partial v_x}{\partial t})}{O(\frac{\mu}{r} \frac{\partial}{\partial r} (r \frac{\partial v_x}{\partial r}))} \sim a^2 \frac{\omega}{\nu}$$

$$a^2 \frac{\omega}{\nu} \ll 1 \Rightarrow 0 = A \cos(\omega t) + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) \Rightarrow v_x = \frac{A \cos(\omega t)}{4\mu} (a^2 - r^2)$$

QUASI-STEADY
POISEUILLE FLOW

$$a^2 \frac{\omega}{\nu} \gg 1 \Rightarrow \rho \frac{\partial v_x}{\partial t} = A \cos(\omega t) \Rightarrow v_x = \frac{A}{\rho \omega} \sin(\omega t)$$

UNIFORM FLOW

THE SOLUTION DOES NOT SATISFY THE BOUNDARY CONDITION ON THE WALL. IN ITS VICINITY, THERE EXISTS A THIN BOUNDARY LAYER WHERE VISCOSITY IS IMPORTANT

$$\rho \frac{\partial v_x}{\partial t} \sim \frac{\mu}{\delta^2} \Rightarrow \delta \sim \sqrt{\nu/\omega}$$

INTRODUCING $\eta = \frac{r-y}{\sqrt{\nu/\omega}}, \frac{A}{\rho \omega} \sin(\omega t) - v_x = \tilde{v}, \omega t = z$

$$\frac{\partial \tilde{v}}{\partial z} = \frac{\partial^2 \tilde{v}}{\partial \eta^2} \quad \begin{cases} \eta=0: \tilde{v} = \frac{A}{\rho \omega} \sin(\omega t) \\ \eta \rightarrow \infty: \tilde{v} \rightarrow 0 \end{cases} \quad \text{STOKES SOLUTION!!}$$

BLOOD FLOW IN THE HUMAN AORTA $\omega \simeq 2\pi \text{ s}^{-1}$, $\nu \simeq 4 \times 10^{-2} \text{ cm}^2/\text{s}$, $a \simeq 1.2 \text{ cm}$

$$a^2 \frac{\omega}{\nu} \simeq 226, \text{ Womersley} \simeq 15$$

VISCOSITY BECOMES IMPORTANT FOR $a \sim \sqrt{\frac{\nu}{\omega}} \sim 0.8 \text{ mm}$ AND IS DOMINANT FOR $a \ll 0.8 \text{ mm}$